

2. Alloys: Al

1. Lightweight metals and aluminium (Al)

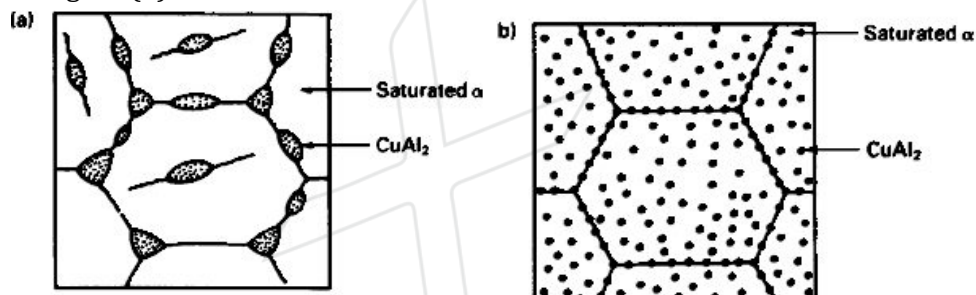
Metals with a density less than 4.5 g/cm^3 at room temperature are called lightweight metals. Commonly used engineering lightweight metals are Al, Mg and Ti.

Pure metals are normally soft therefore it is necessary to strengthen the lightweight metals. The main reinforcing mechanisms are:

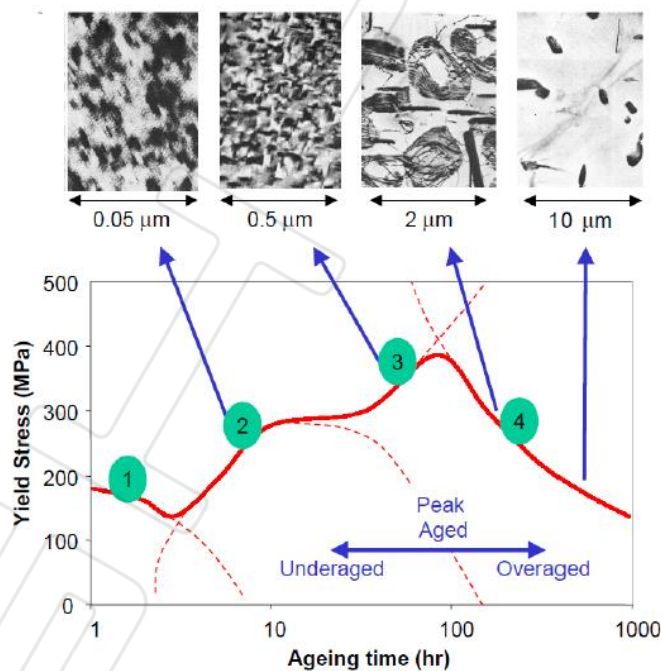
solid solution strengthening, precipitation strengthening, grain refinement and work hardening.

2. Al structure properties relations during heat treatment.

In Figure (a) below, large CuAl_2 particles form at grain boundaries because the Al-Cu alloy is cooled down slowly. This is good/not good for strengthening because rather numerous, small, closely spaced CuAl_2 particle distribution throughout $\alpha\text{-Al}$ grains as in Figure (b) are desired for good strengthening. Therefore we need a method of producing the microstructure as in Figure (b).



Answer questions according to the below figure.



When Al-4 wt% Cu was quenched to room temperature, Super - Saturated - Solid - Solution (SSSS) of α containing Cu can be formed as labelled 1 in the figure above, in which Cu cannot diffuse to form CuAl_2 compound.

In order to obtain the most effective precipitating strengthen effect, a heat treatment (the SSSS will be heated to 100-150 ° C) called tempering is performed.

In Stage 2, Guinier - Preston (GP) zones (disk shape) are formed. The yield stress increases due to coherency stress strengthening.

In stage 3, θ'' phase (metastable phase) is formed. The yield stress increases due to the precipitation strengthening. For this mechanism, when there are numerous, close-spacing and small particles, the particles are rather cut by dislocations; once they coarsen and become larger, fewer and large-spacing particles, dislocation would rather bow-around the particles. Therefore there is an optimum particle size.

In stage 4, θ' phase and θ -CuAl₂ are formed. But now the material is overaged.

Therefore, "peak aging" occurs if we stop the transformation at θ'' phase stage.

3. Naming convention Al

Wrought Al alloys have 4 digits; while cast Al alloys are named as XXXX.

The commonly used Al alloys for aero-applications are:

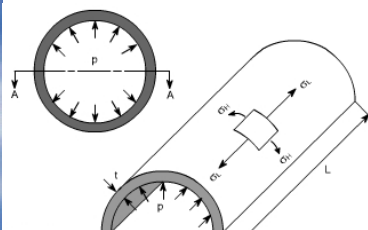
2xxx (main alloy element is Cu);

6xxx (main alloy element is Mg and Si);

7xxx (main alloy element is Zn);

8xxx (main alloy element is Li).

4. Al material selection for planes.



A wing is essentially a beam that is bended during flight. So the upper wing is loaded in compression and cyclic loading, for which compressive strength, stiffness and fatigue resistance should be optimise (normally 7 xxx is chosen). The lower wing is loaded in tension and cyclic loading, so tensile strength, fatigue resistance and fracture toughness should be optimise (normally 2 xxx is chosen).

A fuselage is essentially a pressure vessel during flight. Stress state of fuselage during flight is:

Axial tensile stress;

Circumferential tensile stress.

Each flight is a pressure loading cycle and the fuselage is a fatigue critical area. So these three factors have to be optimise for material selection.

Operating temperature is also important for planes with high speed.

Friction heating can require the aircraft skin to operate at elevated temperature. Concorde (Mach 1.8) was about the limit for

Al alloys. Above mach 2, skin temperatures become too hot for Al alloys and Ti alloys have to be used instead.